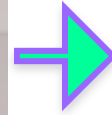
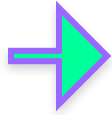


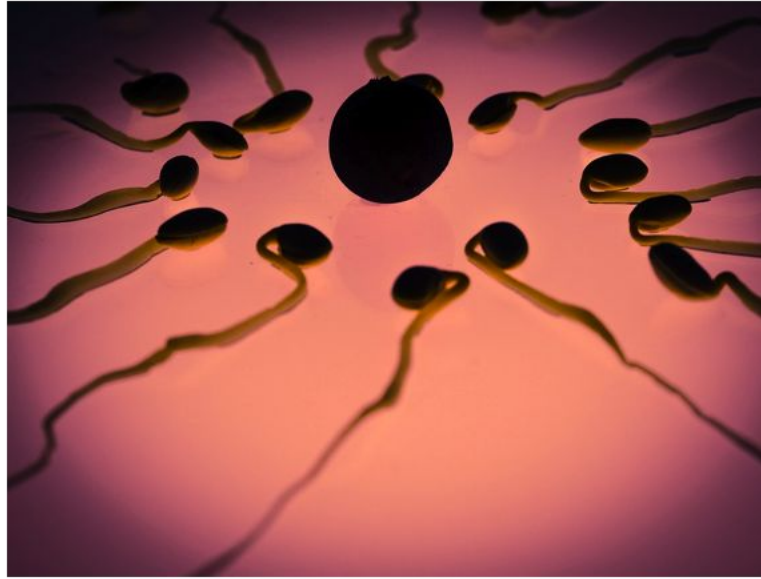
Cell Cycle and Cancer

CONCEPT 3 NOTES

How does this happen?



It all begins with a fertilized egg



- Once the sperm meets the egg and becomes fertilized, that fertilized egg (zygote) goes through cell division repeatedly.
- **Cell division** can give rise to many *identical* cells
- **Differentiation**: a process that creates **special structures and functions**
 - Specialized cells become tissues □ organs □ organ systems.

Complexity of Organisms

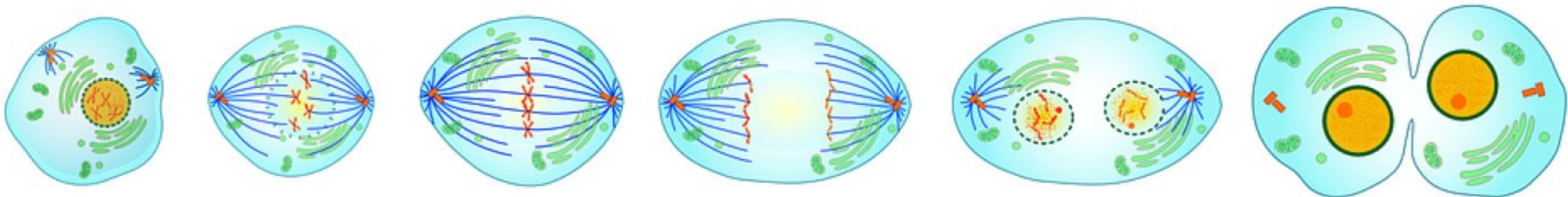
- **Organism**: one individual; one member of a species
- **Organ System**: a distinct set of organs within an organism that work together as a unit for a common function (*ex. Digestive system*)
- **Organ**: several types of tissue that work together for a common function (*ex. stomach*)
- **Tissue**: a group of identical cells working together for a common function (*ex. muscle tissue*)
- **Cell**: most basic unit of life that has all 6 characteristics of living things (*ex. muscle cell*)

Cell Differentiation

- Differentiation is *irreversible*, BUT...
- **Stem cells** = cells that can become differentiated into one or more types of specialized cells.
 - Two Types:
 - **Embryonic Stem Cells**: cells that have never differentiated
 - **Adult Stem Cells**: cells found in adult bone marrow that are partially differentiated and can become bone, blood, cartilage, fat, and connective tissue

Cell Cycle

- **Cell Cycle** - a repeated pattern of growth, DNA duplication and cell division that occurs in EUKARYOTIC cells.
- **2 purposes = GROWTH and REPAIR**
- The Cell Cycle consists of 3 main phases:
 - **Interphase** – cell growth
 - **Mitosis**
 - **Cytokinesis**– cell division which begins at the end of mitosis

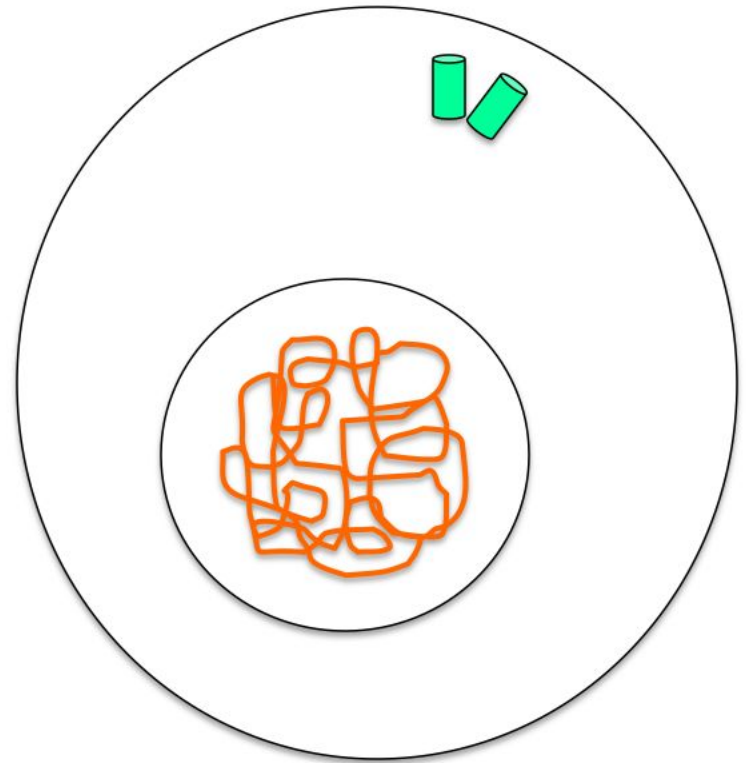


Cell Cycle

Interphase = growth phase of cell. It is divided into 3 phases:

- **G1 (gap 1) phase** - the cell grows and makes proteins.
- **S (synthesis) phase** - chromosomes replicate
- **G2 (gap 2) phase** - cells continue to grow and make proteins

*****At the end of interphase the cell has 2 full sets of chromosomes*****



DNA

- **Chromosome**— one long continuous thread of DNA that consists of numerous genes and regulatory information.
 - Each of your body cells have 46 chromosomes.
 - In normal body cells, chromosomes always occur in pairs



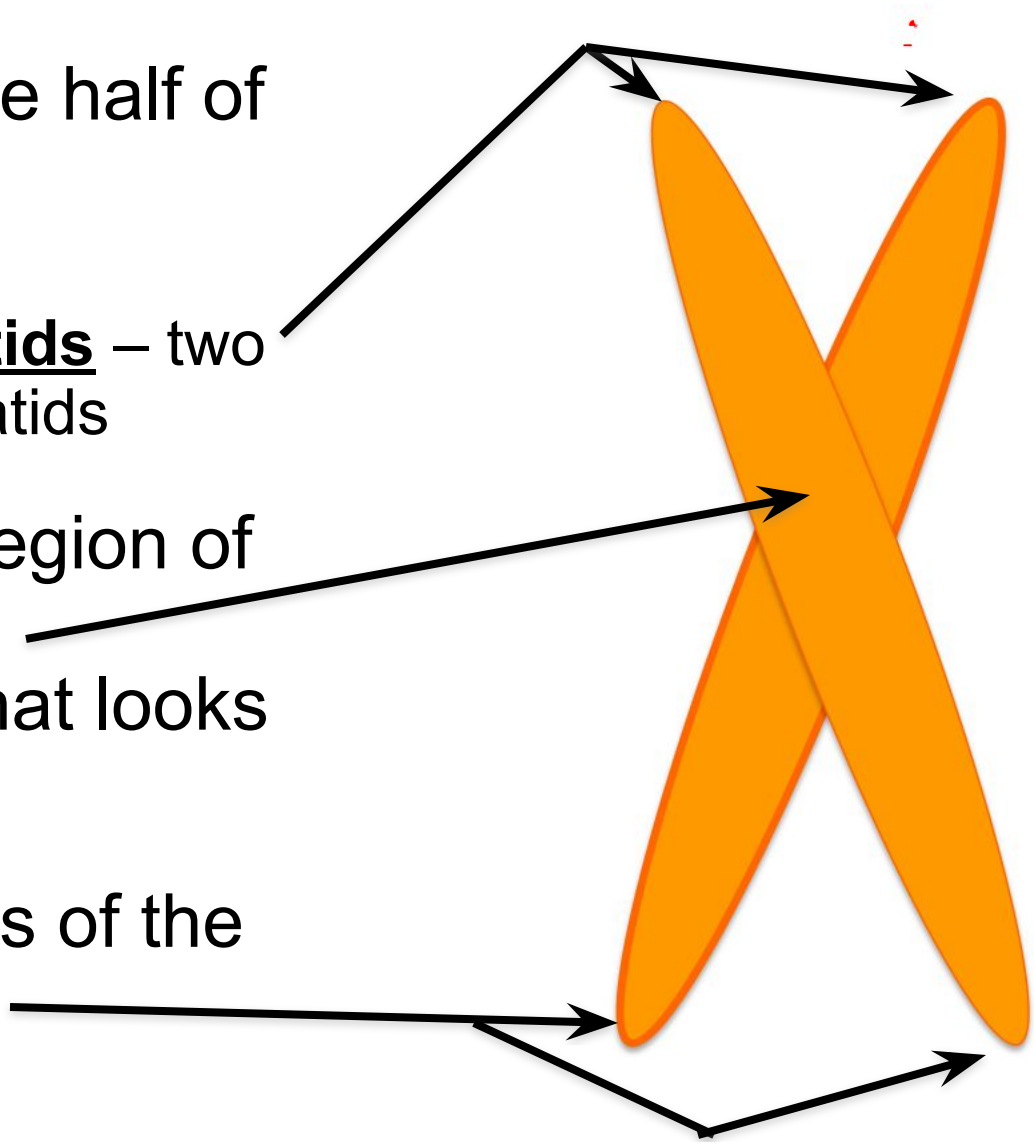
DNA vocabulary:

- **Chromatid** – one half of a duplicated chromosome.

- **Sister Chromatids** – two identical chromatids

- **Centromere** – region of the condensed chromosomes that looks pinched.

- **Telomere** – ends of the DNA molecule.

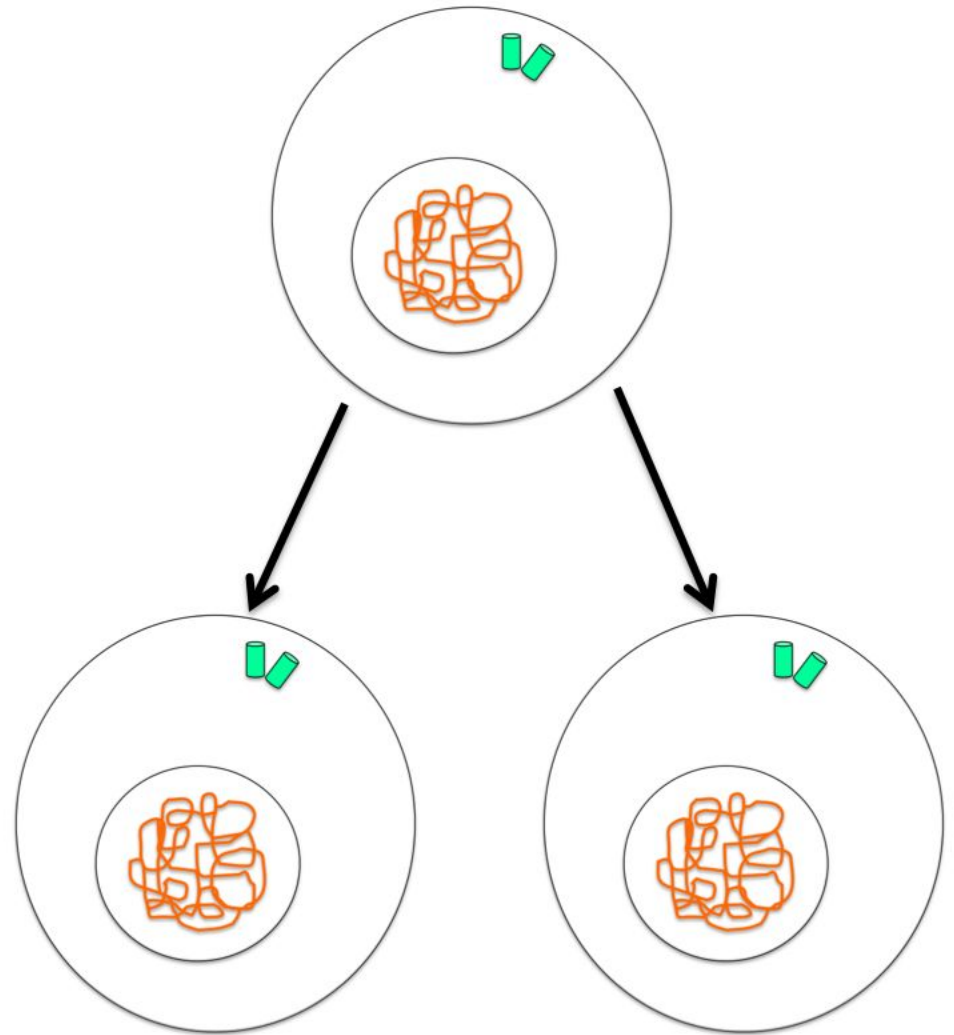


After Cell Preparation during
Interphase, the Cell is ready
to ***DIVIDE***

Mitosis

Mitosis = cell
division

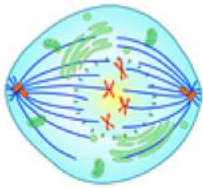
- **1** cell becomes **2**
identical daughter
cells in this phase



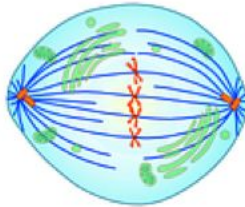
Mitosis and Cytokinesis



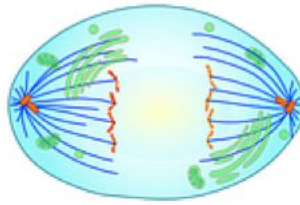
Prophase



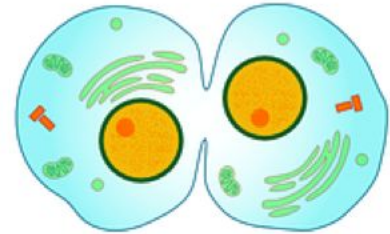
Metaphase



Anaphase



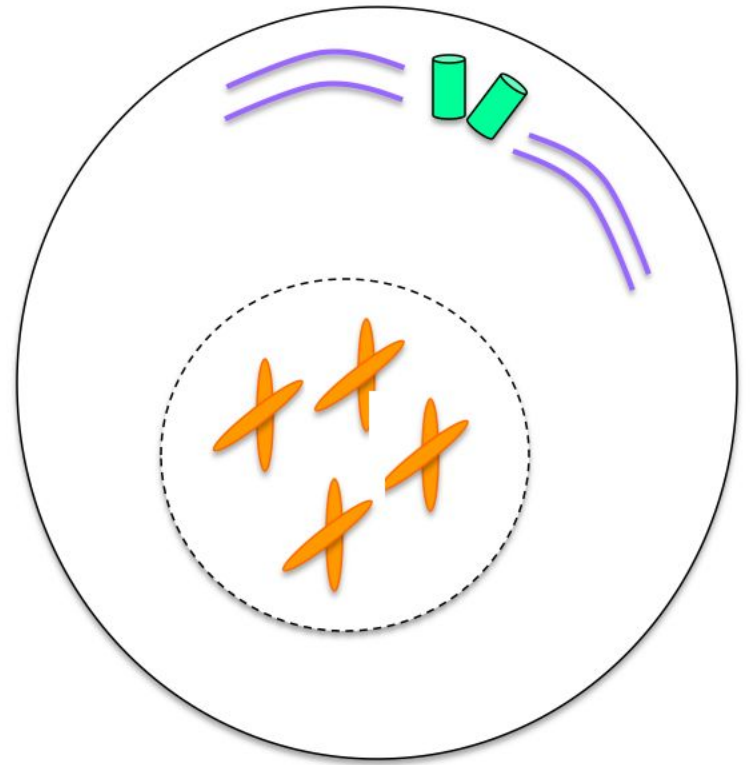
Telophase



Cytokinesis

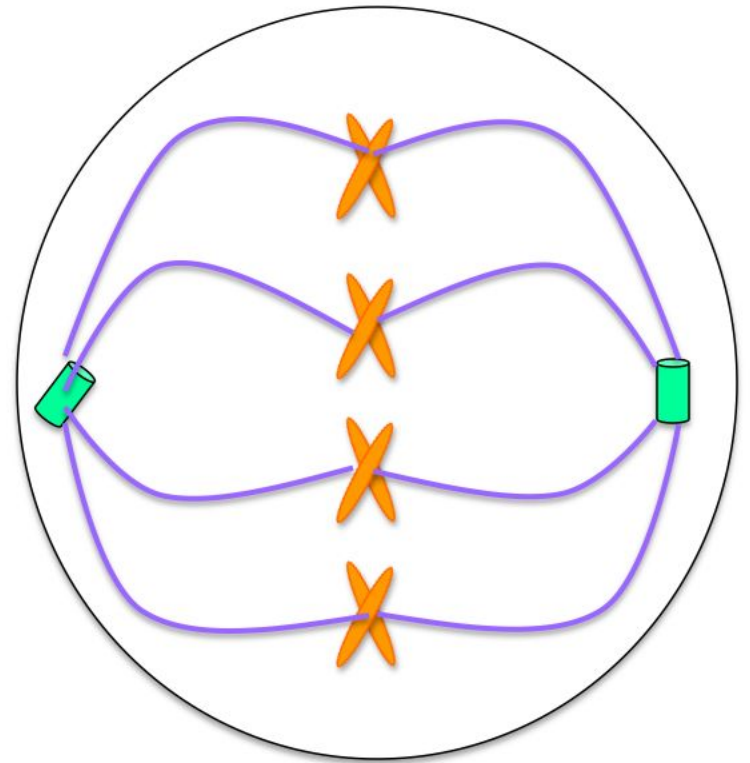
Prophase

- Chromosomes condense & are visible as ***sister chromatids*** (in X's)
- Nuclear membrane disappears.
- Spindle fibers form out of centrioles



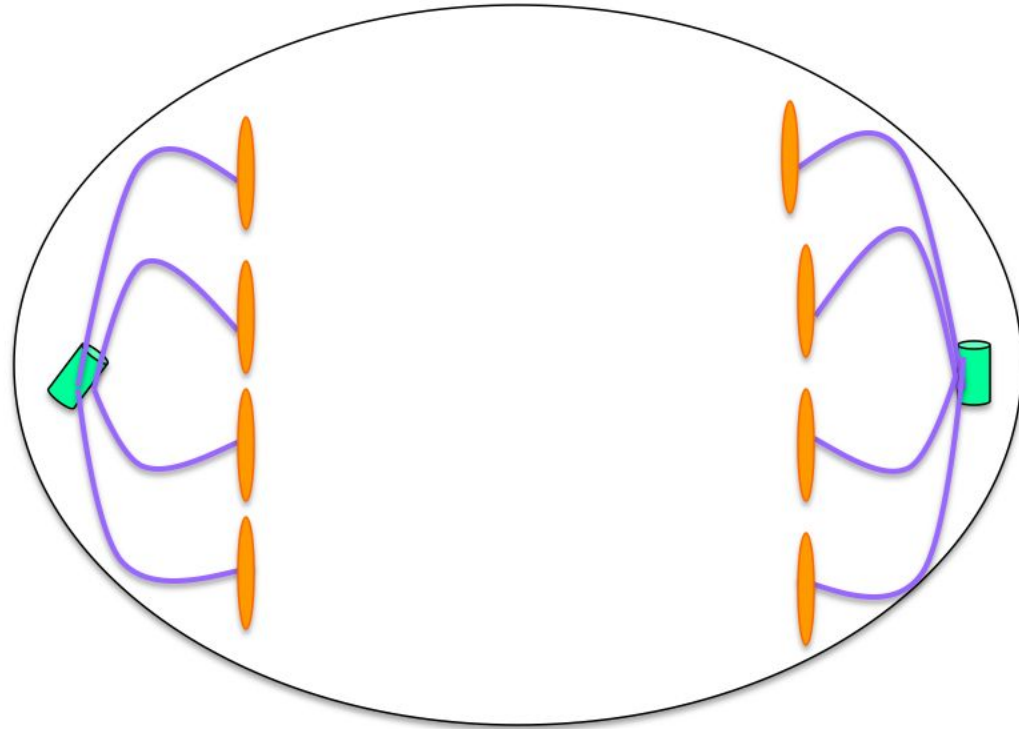
Metaphase

- Spindle fibers connect to the *centromere* of each *sister chromatid*
- Chromosomes ***move to middle*** of the cell



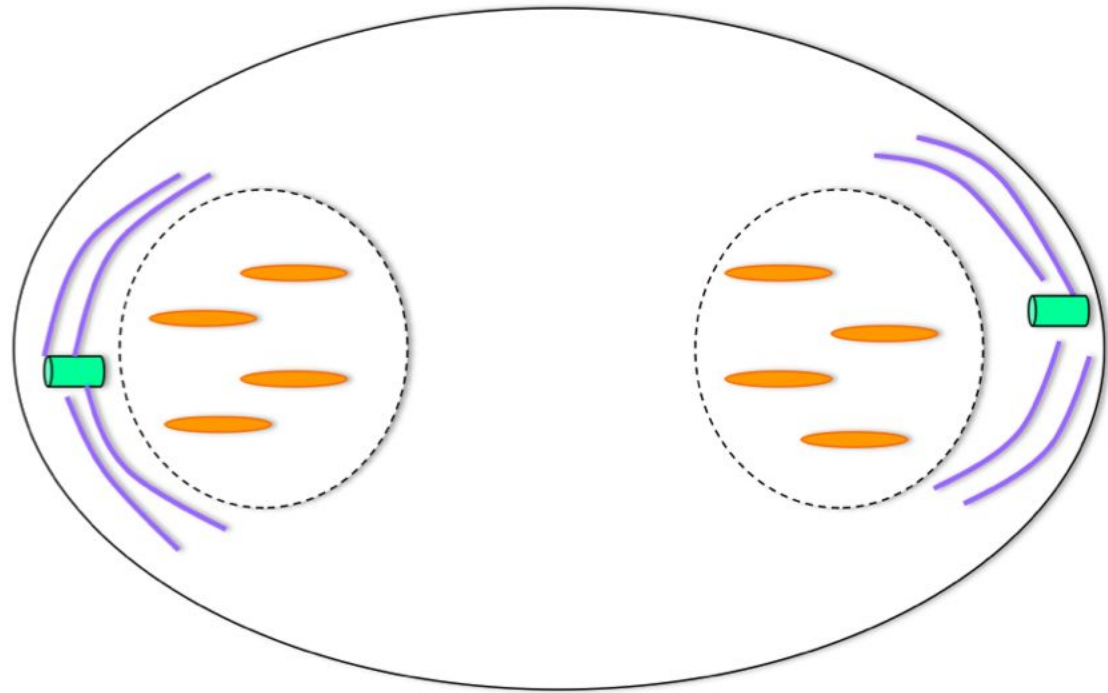
Anaphase

- ***Sister chromatids separate*** becoming individual chromosomes as chromatids move to opposite ends of the cell



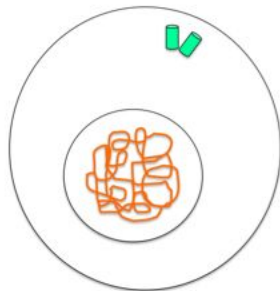
Telophase

- Nuclear envelope forms around chromosomes at each side of the cell.
- Chromosomes start to look like **chromatin** again (threadlike instead of rodlike)
- Spindle fibers break down & dissolve.
- Cytokinesis begins.

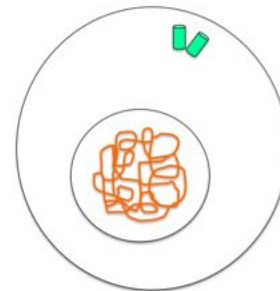


Cytokinesis

- The division of the cytoplasm into two individual cells.
- In **plant cells** – **cell plate** forms midway between divided nuclei & gradually develops into a membrane.
- In **animal cells** - forms a **cleavage furrow** that pinches the cell into 2 equal parts.



End result = 2
identical body
cells



How often do cells divide?

- Every cell divides at a different rate based on its need.
- Examples:
 - Internal lining of intestines: 5 days
 - Skin cells: every 2 weeks
 - Red blood cells: 4 months
 - Liver cells: 1 year

Why do body cells divide?

****Growth and repair****

- Cells can't just get bigger to grow either – they have to stay small to increase surface area and allow things to move in and out easier.

Regulation of the Cell Cycle

How do cells know when to divide
and when not to?

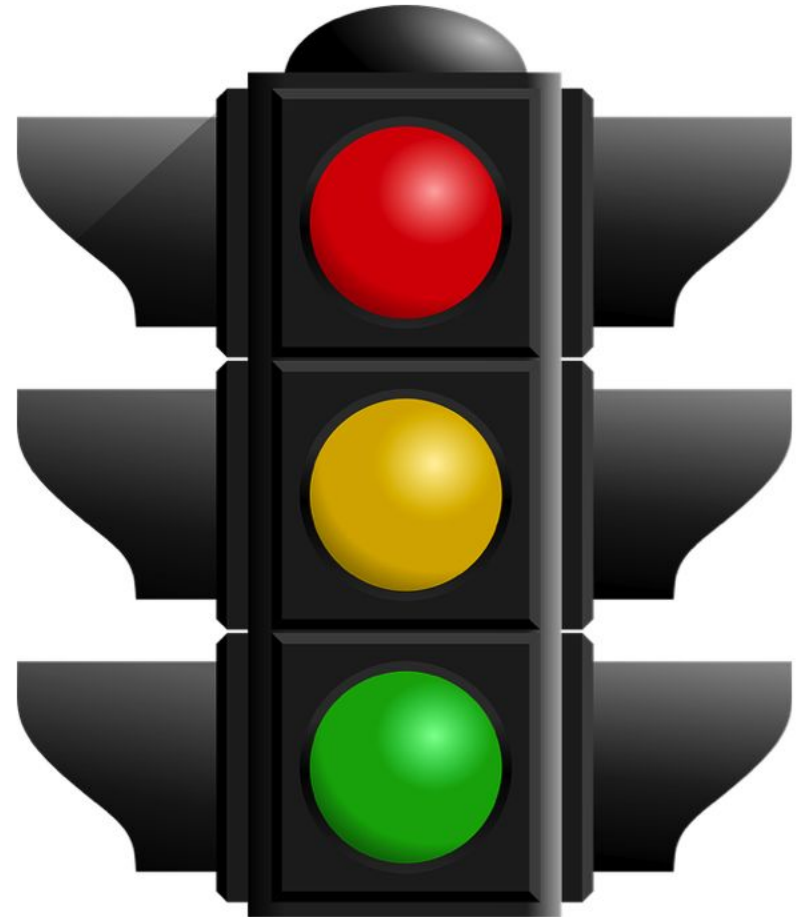
Regulation of the Cell Cycle

Cell cycle is controlled by a *chemical control system* that starts & stops events in the cell cycle.

1. External Regulation— signal that comes from outside of the cell; *Ex. Hormone, nutrients, etc.*
2. Internal Regulation— signal that comes from the nucleus; *Ex. DNA inside of the cell*

Checkpoints

- **Checkpoint** – critical point where “stop” & “go” signals can regulate the cycle.
 - *Note: Cell division is mostly in the “off” position in animal cells if there is no stimulus present.*



Apoptosis

- Apoptosis – programmed cell death.
 - Internal / external signals activate genes that produce self-destructive enzymes.
 - Nucleus shrinks and breaks apart.
 - **Ex:** Web fingers & toes during development



Cancer

- Cancer – uncontrolled cell division
 - Happens when the regulation of the cell cycle breaks down.
 - Cancer cells divide much more often than healthy cells do.
 - Leads to the formation of tumors
- Tumors – clumps of cells that divide uncontrollably
 - Two types: Malignant and Benign

Tumors

- Malignant – cancer cells can breakaway from the tumor & be carried to other parts of the body/organs & form more tumors
 - Metastasis = spread of disease from one organ to another.
- Benign – abnormal cells typically remain clustered together.
 - It may be harmless & easily removed



Causes

- Often the result of exposure to carcinogens = cancer causing agents (chemicals that can cause cancer)
 - *Ex. UV rays, tobacco smoke, X-rays, etc.*
- Carcinogens mutate DNA
- Other random mutations and some infectious agents can also cause cancer (*Ex. HPV*)

